

Database Management

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Data stored in File systems:

- issued_books.txt : (Jan)

1. Issue-101, DBMS-Korth, Ravi, CSE, 208321 10-01-2026 , 25- 01-2026
2. Issue-102, DBMS-CJDate, Anu, ECE, 208323,12-01-2026, 27-01-2026
3. Issue-102, OS-Silberschatz, Anu, ECE, 208323,12,12-01-2026, 27-01-2026

What are the issues?

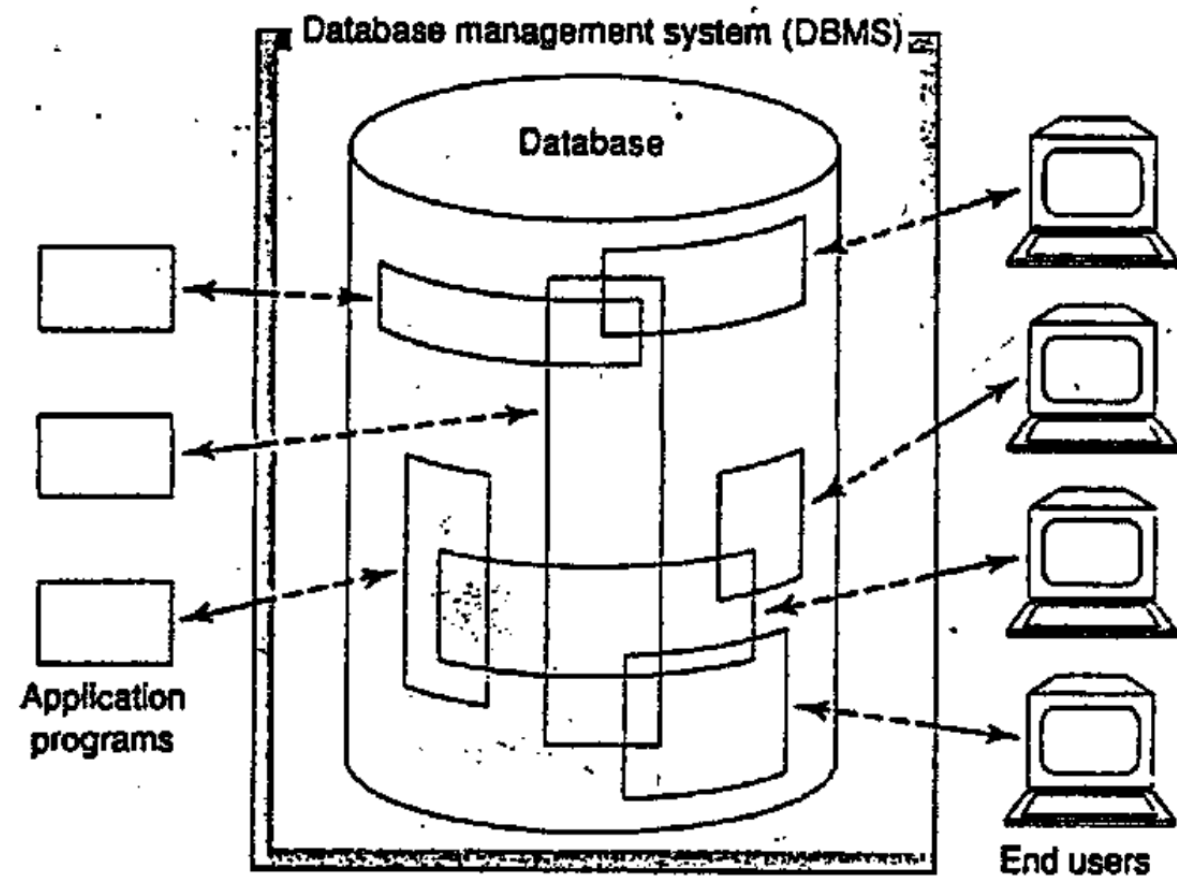
Common pitfalls in file systems

- Student details/Book details repeated (**Data redundancy**)
- No validation on multiple books issue (**Data validation**)
- Multiple formats (eg - date format mm-dd-yy, dd-mm-yyyy) of entries by multiple users - (**Data inconsistency**)
- Scattered across multiple files - (**Data isolation**)
- No validation on retrieved data (**Data integrity**)
- Speed (**Sequential entries**)
- Security (**Who can add the entries?**)

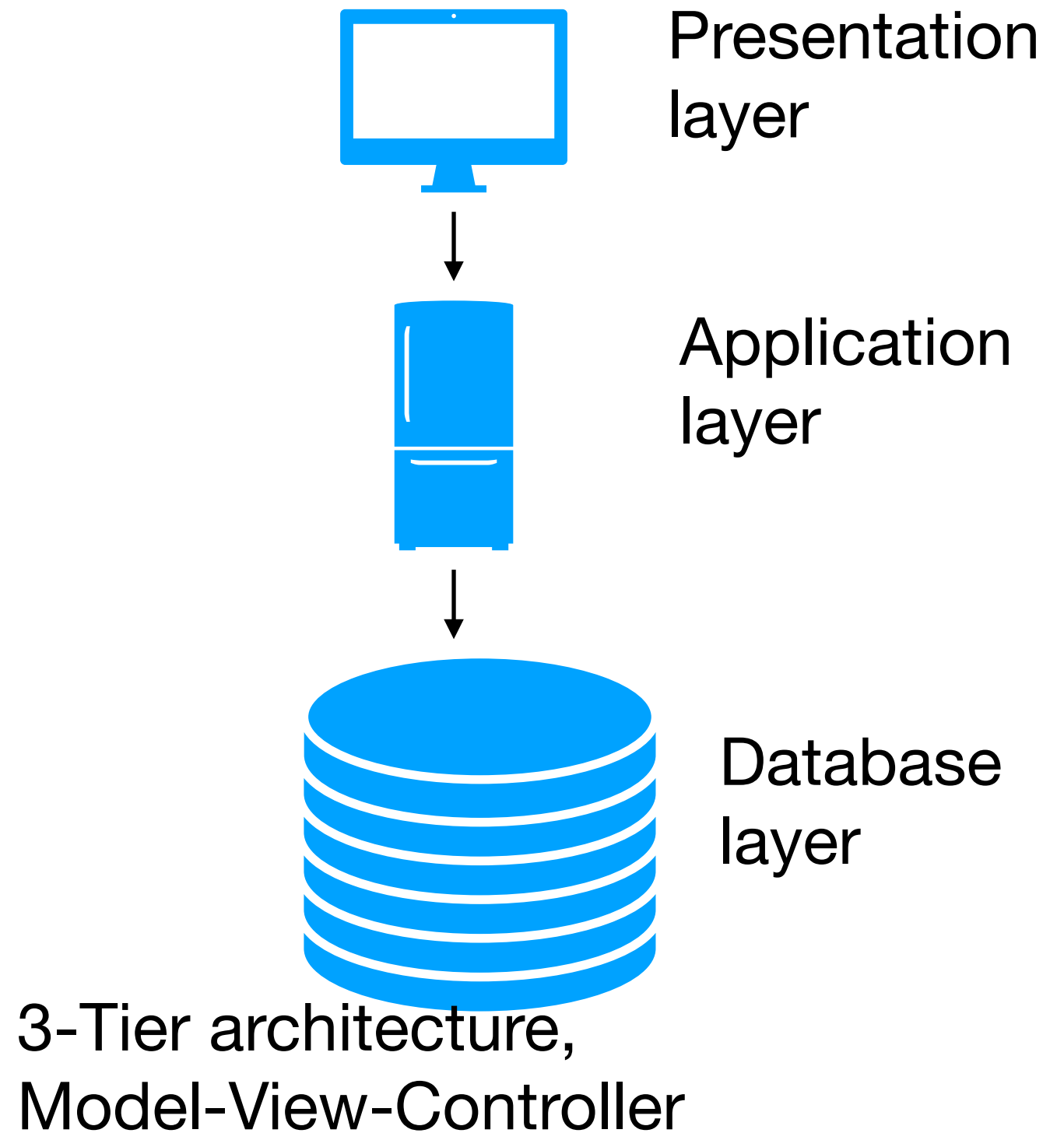
What is a DBMS?

- Computerized system to store data and allow users to retrieve and update information on demand
- Hardware - Software - Users - Data
- Integrated storage (multiple files)
- Shared access (Different users)
- Concurrent access (Multiple users - Same time)

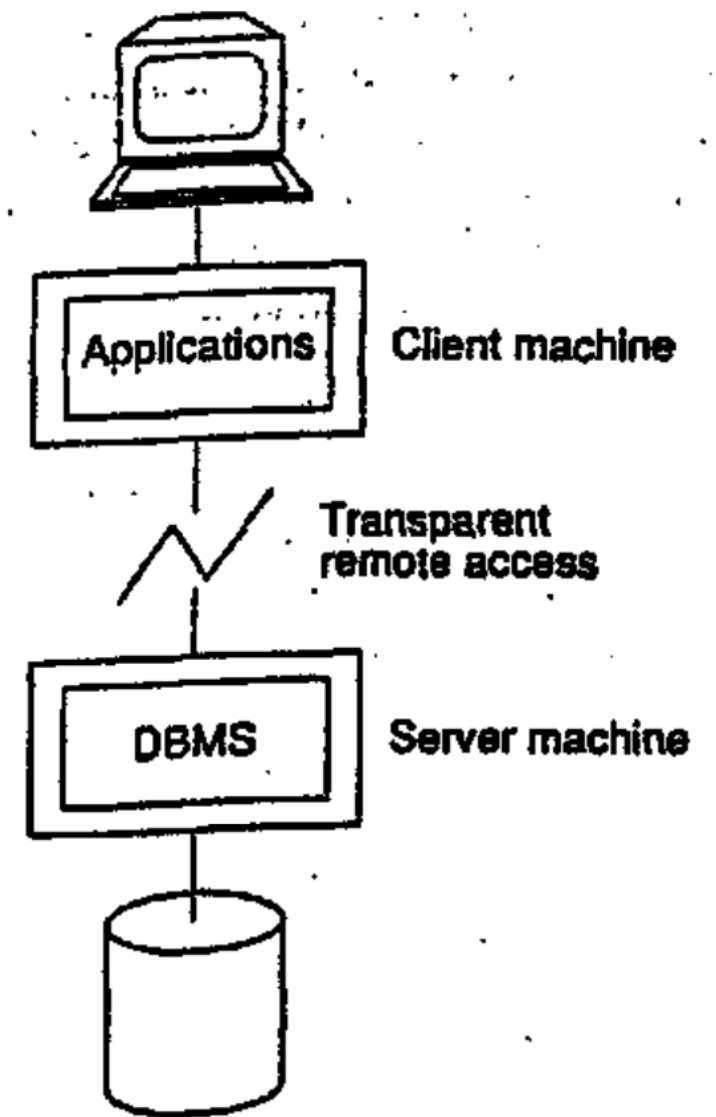
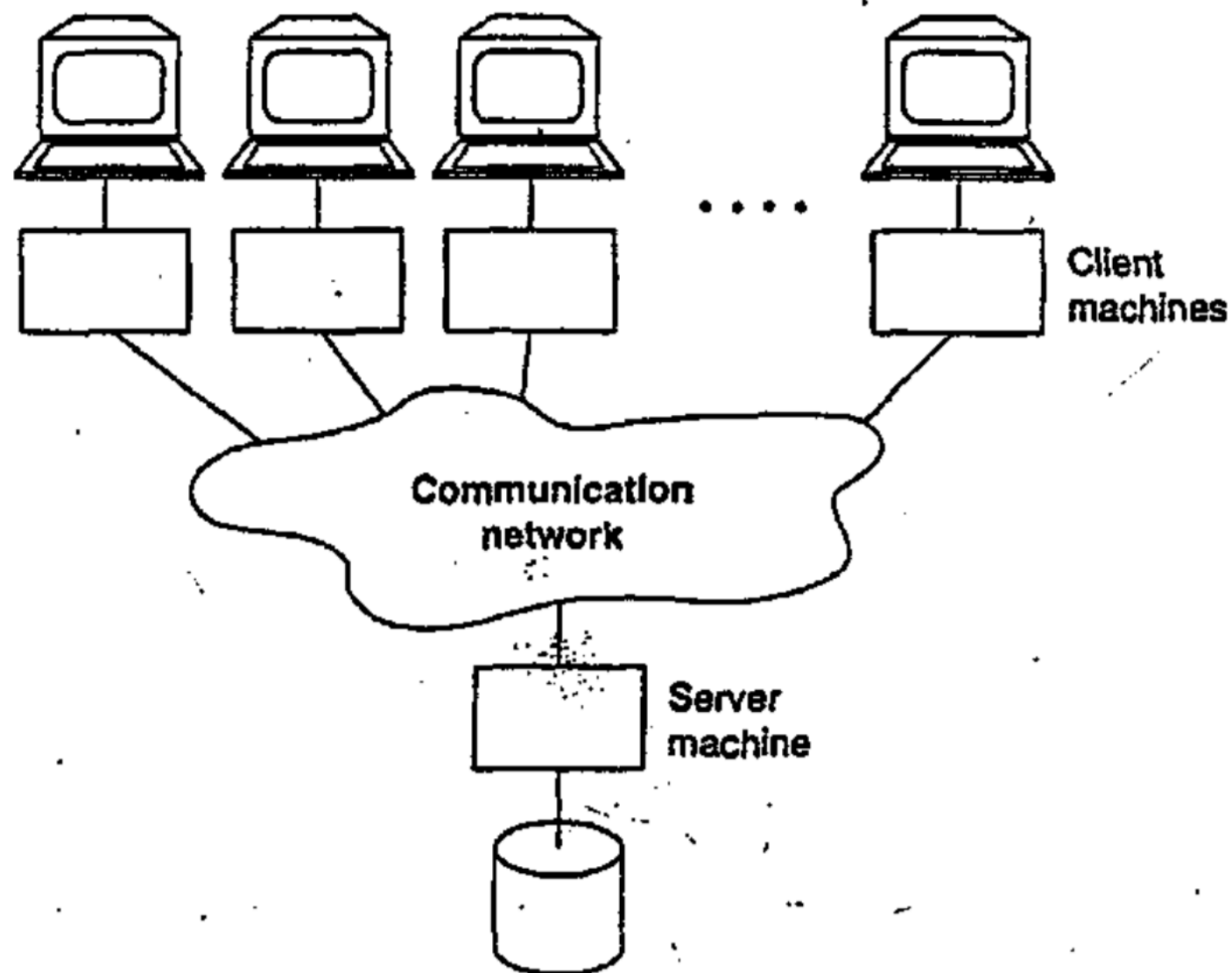
Database Management Systems



Adapted from CJ Date



Distributed architectures



Components

- Hardware - Magnetic disks , Solid State Drives, NVMe SSD - Query Language
- Software - Oracle Database, IBM DB2, MySQL, Microsoft SQL Server, PostgreSQL,
- Users - application programmers (web app developers), Database administrators, end-users
- Data - Information records (rows in tables)
- SQL - Structured Query Language

Relational DB - example

books

Book-id	Book-name	Author	ISBN
IBAB-BDB101	Database Basics	David Korth	SBN-13 978-0-323-77671-4
IBAB-BDB102	Operating Systems	Andrew Tanenbaum	ISBN-13 876-0-323-77634-2
IBAB-BDB103	Database Systems	CJ Date	ISBN-13 896-0-543-74435-5

students

Student-id	Student-name	Department	DOB	Email
208321	Ravi Vytla	CSE	22-05-1990	208321@ibab.ac.in
208323	Anu Hamsa	ECE	12-11-1990	208323@ibab.ac.in

Each row
—> record

Enforce constraint
- Only one book
per user

Issue-id	Book-id	Student-id	Issue-date	Return-date	Returned
IBAB-012026-1	IBAB-BDB101	208321	10-01-2026	25-01-2026	Yes
IBAB-012026-2	IBAB-BDB103	208323	12-01-2026	27-01-2026	No

book_issues

Database Languages

- Data Definition Language (DDL) :

```
CREATE TABLE STUDENTS (  
    Student-id INT PRIMARY KEY,  
    Student-name VARCHAR(50) NOT NULL,  
    Department VARCHAR(30),  
    Email VARCHAR(100) UNIQUE,  
    DateOfBirth DATE  
);
```

```
ALTER TABLE STUDENTS  
ADD Phone VARCHAR(15);
```

```
DROP TABLE STUDENTS;
```

Queries

- Data Manipulation Language (DML) - Query Language

```
SELECT B.Book-name FROM BOOKS B  
JOIN ISSUES I ON B.Book-id = I.Book-id  
WHERE I.Student-id = 208323;
```

```
SELECT * from BOOKS;
```

```
SELECT * from STUDENTS;
```

DDL vs DML

- DDL - Create, update, alter , truncate, rename, drop
- DML - select, update, insert, delete, merge
- Check documentation for syntax : <https://dev.mysql.com/doc/refman/8.4/en/sql-data-manipulation-statements.html>

Users

- Web-app developers (Programmers of Moodle web app)
- End-users (students using Moodle LMS)
- Database administrators (Design the database)
 - Schema definition
 - Storage,
 - maintenance
 - Authorization

What is different across different Database servers?

- SQL syntax (mostly DDL)
- Datatypes (boolean, date/time)
- Constraints (Not null, Primary Key, Foreign Key)
- Transactions and isolation levels
- Stored Procedures/Functions (IF-ELSE blocks, return values) - eg. PL/SQL for Oracle, T-SQL for SQL Server

Primary Key/Foreign Key

- **Primary Key** - Unique, Not Null - identifier for a record, can be a single column or a combination of columns
- **Foreign Key** - column in Table B that was a primary key in table A - to identify relationships, need not be not null/unique

Normalization

- Organize data in tables such that there is no redundancy and avoid data inconsistency.
- First Normal Form - No multiple values allowed, Primary Key needed
- Second Normal Form - remove redundancy by splitting - No partial dependency
- Third Normal form - remove transitive dependencies

History of Database systems - Interesting video

- <https://www.youtube.com/watch?v=6szdySvorzA&t=552s>

Entity-Relationship Diagrams

- Entity - Weak, Strong (Primary Key, Partial Key)
- Attributes - Key, Composite, Derived, Multi valued
- Relationship degree - Unary, Binary, Ternary, N-ary
- Cardinality - (1:N, N:1, 1:1, N:N)